Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/11/2019
PI Name:	Smith, David J Ph.D.		
Project Title:	Accumulated Doses and Microbial Mutagenesis from Ionizing Radiation Exposures (ADMMIRE) over Antarctica		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	10/01/2018	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gentry, Diana Ph.D. (NASA Ames Research Center) Venkateswaran, Kasthuri Ph.D. (Jet Propulsion Laboratory)		
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Task Description:	How bacteria respond to low dose rate (LDR) ionizing radiation is poorly understood because such experiments are often impractical at ground-based research facilities. Alarmingly, previous spaceflight studies have hinted at evolved bacterial virulence and emerging novel species in spacecraft. It is conceivable that the radiation quality in space, including high energy ions and protons, were responsible for these observations; however, orbital spaceflight experiments include both microgravity and high energy radiation exposures (simultaneously) making it hard to untangle the influence of each stressor. Balloons launched to the polar stratosphere can be used to perform studies capable of measuring the influence of LDR ionizing radiation exposures, alone, on microorganisms. Our study, "Accumulated Doses and Microbial Mutagenesis from Ionizing Radiation Exposures (ADMIRRE) over Antarctica," will use the Exposing Microorganisms in the Stratosphere (E-MIST) payload flown on a long duration polar balloon mission. Previous surveys of the International Space Station (ISS) have revealed that most viable environmental bacteria are dormant, accumulating radiation over time but not actively growing on surfaces. How does prolonged dormancy in the space radiation environment impact the behavior of bacteria when opportunistically resuscitated? Our proposed experiment can address this major unknown with four achievable objectives: (1) Prepare the E-MIST payload and bacterial spores for a long-duration balloon flight; (2) Fly the experiment to the Antarctic stratosphere and measure environmental conditions; (3) Determine the effect of near space polar radiation on bacterial samples; and (4) Simulate near space radiation conditions using ground-based facilities. Our overarching project hypothesis is that accumulated LDR ionizing radiation will result in measurable, non-lethal mutations in dormant Bacillus siamensis IIF7SW-P1 spores. The E-MIST payload was designed by our team and has flown multiple times to the stratosphere as a NASA piggyback payload, including an Antarctic flight launched in December 2018. For this proposed experiment, we will work with ISS-isolated bacteria collected from Space Biology's Microbial Tracking-1 project. The bacterial specimens will be sporulated to create dormant, uniform, desiccated and stable samples; thus, our strains are ideal for Antarctic mission logistics that entail long handover and retrieval times. Pre-flight ground chamber testing, hardware refurbishment, and instrument calibrations will be done at Kennedy Space Center (KSC) by members of our team. The E-MIST payload will include a Technology Readiness Level-9 (TRL-9) dosimeter for measuring the radiation environment; additionally, we will insert shielding (aluminum, titanium, and polypropylene) with varying thicknesses 'upstream' of bacterial sample plates in order to determine the effect of radiation absorbance and/or particle scattering. Pre- and post-flight microbiological processes will be done at Ames Research Center (ARC) and Jet Propulsion Laboratory (JPL), where surviving bacteria will be germinated and characterized using state-of-the-art molecular assays. Project omics data (RNASeq differential expression analysis) will be archived in the NASA GeneLab repository. By studying the response and potential mutagenesis of dormant bacteria in space-like radiation conditions, our project is directly aligned with priorities outlined in the Space Biology Science Plan 2016-2025 and Decadal Survey, specifically: MB-1 (What underlying genetic, molecular and biochemical processes are influenced by the space environment?) /Decadal P1 (multigenerational studies of microbial populations), P2 (analysis in space of plant and microbial growth and of the physiological responses of plants and microbes to the spaceflight environment) and MB-6 (Does long-duration spaceflight alter normal rates of evolutionary change?) /Decadal P1. Using the NASA balloons to deliver a natural, complex, LDR ionizing radiation exposure in the polar stratosphere will allow our team to better understand how spacecraft-associated bacteria could be influenced by radiation levels expected on a long duration mission to Mars, and if any changes pose a potential risk to astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project seeks to distinguish between acute and LDR radiation exposure effects on bacterial spores, with measurements of antibiotic resistance (and associated genes) in surviving specimens. Antibiotic resistance is an growing public health threat on Earth. A deeper understanding of antibiotic resistance profiles -- regardless of the trigger (in our case: ionizing radiation) -- will be useful to the scientific community searching for solutions.
	Due to delays in first ADMMIRE balloon flight opportunity in Antarctica, we have been largely on “pause” and operating at a very low level of effort in the first year performance period – in order to preserve grant resources (procurement/travel) for flight years. This decision was made after discussions with Balloon Program Office in early 2019, who gave indications that it might be 2-3 years before E-MIST could fly again from Antarctica for the ADMMIRE project. To date, labor at Ames Research Center (ARC) and Kennedy Space Center (KSC) has been used to keep the team intact and to make some advances in the payload design, particularly with new radiometers and dosimeters. Dr. Ye Zhang at KSC has been kept in the loop on our request to extend the ADMMIRE grant performance period to 9/30/2022. Some noteworthy milestones: • July 2018: Antarctic LDB Flight Application for FY21-26 period submitted to Balloon Program Office and Columbia Scientific Balloon Facility (CSBF). Awaiting flight assignment. • Dec 2018: First E-MIST payload launched from Antarctica (for Planetary Protection grant) in FY18. Balloon did not reach altitude due to anomaly. • Jan 2019: Payload recovered from the ice. • March 2019: Payload returned to KSC for refurbishment. • June 2019: Environmental permits extended for both New Zealand and Antarctica. Will remain valid through 2023. • July 2019: E-MIST payload integration at CSBF (for Planetary Protection grant). • August 2019 (anticipated): Integration & compatibility testing at CSBF for Antarctica FY19 flight opportunity (for Planetary Protection grant). • September 2019 (anticipated): First test of dosimeters and radiometer to be used for ADMMIRE project on FY19 LDB test flight from Ft. Sumner. • December 2019 (anticipated): Second Antarctic flight opportunity for E-MIST (for Planetary Protection grant). • March 2020 (anticipated): Conclusion of Planetary Protection grant and refurbishment of E-MIST payload use on future ADMMIRE balloon flights.

Task Progress:	Task 1: Refurbishment and testing of E-MIST hardware & instruments • Payload refurbished at KSC after Dec 2018 launch from Antarctica for other research grant (Planetary Protection). • Payload at Columbia Scientific Balloon Facility (CSBF) in Palestine, Texas, July 10-12 for environmental chamber testing, prior to next Antarctic launch attempt for other research grant (Planetary Protection). Task 2: Machine coupons of varying thicknesses for carrying spores • Aluminum coupons are ready ; • Polypropylene and titanium coupons have not yet been manufactured Task 3: Prepare bacterial spores • Bacillus siamensis strains from MT-1 are available and ready for use ; • Protocol for spore prep has been established and finalized ; • Scanning electron microscope methods have been tested and validated on other bacterial spores (Bacillus sp.) ; • Environmental permit for ACA and transship through New Zealand both extended through March 2023 for future ADMMIRE flight Task 4: Maintain ground and transport controls • Bacillus siamensis controls will be produced prior to ground and flight experiments Task 5: Record environmental conditions and radiation doses • We have made major strides in improving the E-MIST payload radiometer and dosimeter capabilities, through citizen scientist collaborations (on the radiometer) and DLR (German Aerospace Center) collaborations (for passive TLD and active M42 dosimeter) • Ultraviolet radiometer measurements tested on smaller (meteorological) weather balloon launched in October 2018 – results published in Aerobiologia (Caro et al., 2019) • New UV-C sensor being developed and tested at NASA Ames for ADMMIRE flight • Dosimeters from DLR collaboration (TLDs and M42) being tested on FY19 Ft. Sumner NASA Balloon Flight (LDB Test Flight) with CSBF in August 2019 Task 6: Survival of bacterial samples • We have been performing x-ray studies in the Aerobiology Lab at Ames to optimize bacterial survival and sequencing assays, in prep for future ground- and flight-based experiments Task 7: Antibiotic resistance tests for spores surviving radiation--Will be performed in upcoming project years Task 8: Differential expression for spores surviving radiation exposures--Will be performed in upcoming project years Task 9: Low LET control studies testing bacterial spore resistance--Will be performed in upcoming project years Task 10: Accelerated iron ion studies testing bacterial spore resistance--Will be performed in upcoming project years. Also pursuing “piggyback” beam time in Chiba, Japan as a way of potentially reducing costs for using NASA Space Radiation Laboratory (NSRL).
Bibliography Type:	Description: (Last Updated: 12/09/2021)
Articles in Peer-reviewed Journals	Caro TA, Wendeln M, Freeland M, Bryan N, Waters SM, McIntyre A, Nicoll P, Madronich S, Smith DJ. "Ultraviolet light measurements (280 to 400 nm) acquired from stratospheric balloon flight to assess influence on bioaerosols." Aerobiologia. 2019 Dec;35(4):771-6. https://doi.org/10.1007/s10453-019-09597-9 [NOTE reported originally in July 2019 as "First online 24 June 2019"] , Dec-2019
Books/Book Chapters	Smith DJ, Waters SM. "Stratosphere Biology." in "Encyclopedia of Astrobiology." Ed. M. Gargaud et al. Berlin, Heidelberg: Springer, 2019. https://doi.org/10.1007/978-3-642-27833-4 , Jun-2019